Jiawei Pei

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Education

University of Zurich — *Sep. 2024 – Jan. 2027*

M.Sc. in Artificial Intelligence, minor in Data Science — *Zurich, Switzerland*

Wuhan University — *Sep. 2019 – Jul. 2023*

B.Eng. in Electronic Information Engineering — *Wuhan, China*

- **Research focus:** LLM agents, post-training, reinforcement learning for tool use, preference optimization, and search-augmented generation.
- **Languages:** English (IELTS 7.5; Duolingo English Test 125).

Profile

M.Sc. student in Artificial Intelligence with hands-on research engineering experience in LLM post-training, agentic reinforcement learning, and evaluation for weak-context generation. I am interested in building language agents that retrieve evidence, use tools, and align responses with user intent through scalable training and evaluation pipelines.

Industry Research Experience

OPPO Guangdong Mobile Communication Co., Ltd. — *Mar. 2026 – Present*

LLM Algorithm Engineer Intern — *Guangdong, China*

Xiaobu Input Method - AI Smart Reply and Personalized Response Generation

Worked on post-training and evaluation for AI reply generation in weak-context messaging scenarios, covering data construction, distillation, preference optimization, and offline evaluation.

- Reframed input-method prediction from word-level completion to short-reply generation using conversation history, user draft, and style tags as structured inputs.
- Built SFT data-cleaning and DPO preference-construction pipelines based on failure cases such as self-questioning, history copying, draft ignoring, and generic replies.
- Implemented post-training experiments with veRL, FSDP, and vLLM on $2\times$A100, covering SFT, 27B-to-4B distillation, GRPO, and weighted-DPO.
- Evaluated 2,000 held-out examples with six-dimensional LLM-as-Judge scoring; the best model reached 4.4243/5 overall score, improving over the original 4B model by 24.9%.

Selected Research Projects

Agentic Search-RL: Reinforcement Learning for Tool-Augmented QA

Jan. 2026 – Mar. 2026

Developed an open-domain QA agent that can actively call a retrieval tool during inference, turning a fixed RAG pipeline into model-decided search actions optimized with PPO.

- Formulated QA as a multi-turn *think* $\rightarrow$ *search* $\rightarrow$ *observation* $\rightarrow$ *answer* process, allowing the model to decide when external evidence is needed.
- Built a local Wikipedia retrieval environment for Qwen2.5-3B rollouts, parsing model-generated search actions and injecting top-k evidence snippets as observations.
- Trained the agent with veRL / PPO on $4\times$A100 using vLLM for rollout; macro-average score improved from 0.0388 to 0.2720 across six QA benchmarks.

DunmyHead: End-to-End Lightweight LLM Training System

Oct. 2025 – Nov. 2025

Reproduced a compact decoder-only LLM training workflow from model implementation to pretraining, instruction tuning, preference alignment, inference, and evaluation.

- Implemented core LLaMA-style modules in PyTorch, including RoPE, RMSNorm, SwiGLU, causal attention, token embeddings, and LM head.
- Built a complete pretraining $\rightarrow$ SFT $\rightarrow$ DPO pipeline with tokenizer, Chinese corpus, instruction data, preference data, checkpointing, and inference tests.
- Improved C-Eval / C-MMLU accuracy from 25.48% to 27.49%; DPO achieved higher subjective QA quality than SFT and pretraining baselines.

Additional Publications and Patent

[J1] Shaolong Wu, **Jiawei Pei**, Yuchen Wu, Wenxuan Shao, Xinting Zheng, Runfan Zhang. "LiDAR Point Cloud-Based Tower-Type Recognition for Overhead Transmission-Line Towers." *Geomatics Science and Technology*, 10(3): 161–172, 2022.

[J2] Guangyi Yang, Songyu Ding, **Jiawei Pei**, Jintang Hu. "Design and Implementation of a Comprehensive Experiment on Mixing Distortion in High-Frequency Circuits." *Experimental Science and Technology*.

[P1] Chi Chen, Shaolong Wu, Zhiye Wang, Bisheng Yang, **Jiawei Pei**, Xinting Zheng, et al. "Method and Apparatus for Transmission-Line Tower-Type Recognition Based on 3D Point Cloud Data." Chinese Patent CN202210908110.2.

Technical Skills

LLM Training	SFT, DPO, weighted-DPO, PPO, GRPO, distillation, preference-data construction, checkpoint management, LLM-as-Judge evaluation.
Systems	PyTorch, veRL, TRL, PEFT, HuggingFace Transformers, FSDP, LoRA, vLLM inference, multi-GPU training, asynchronous rollout.
Agents and RAG	ReAct, tool calling, search agents, retrieval-augmented generation, evidence injection, multi-turn retrieval, answer extraction.
Programming	Python, C++, SQL, Linux, Git, Docker, tmux, ModelScope, SwanLab, Weights & Biases, Claude Code, Codex.